

CEREBROSPINAL FLUID FLOW

The tissue of the brain floats in cerebrospinal fluid (CSF) within the skull. CSF is a clear liquid, which is renewed four to five times a day. CSF protects and nourishes the brain and spinal cord as it flows around them. It contains proteins and glucose that provide energy for brain cell function, as well as

lymphocytes that guard against infection. CSF is produced by the choroid plexuses in the lateral ventricles, and drains into the third ventricle. It then flows into the fourth ventricle, which is located in front of the cerebellum. Circulation of the fluid is aided by pulsations of the cerebral arteries.

1 Site of fluid production (choroid plexuses)

The CSF found in the ventricles in the brain is produced in clusters of thin-walled capillaries, known as choroid plexuses. These capillaries line the walls of the ventricles.

4 Site of reabsorption (arachnoid granulations)

After circulating around the brain, CSF is reabsorbed into the blood via structures known as arachnoid granulations, which are projections of the arachnoid layer into the large sagittal sinus, or cerebral vein.

2 Direction of flow

Fluid moves from the brain's lateral ventricles into the third and fourth ventricles. The fluid then flows up the back of the brain, down around the spinal cord, and up to the front of the brain, as indicated by the arrows.

Lateral ventricle

Dura mater

Third ventricle
Cerebellum

Fourth ventricle

Spinal cord

Central canal

Skull

3 Circulation around spinal cord

CSF ebbs and flows around the spinal cord and through its central canal.

